

Double machine learning-based programme evaluation under unconfoundedness

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Summary: This paper reviews, applies, and extends recently proposed methods based on double machine learning (DML) with a focus on programme evaluation under unconfoundedness. DML-based methods leverage flexible prediction models to adjust for confounding variables in the estimation of (a) standard average effects, (b) different forms of heterogeneous effects, and (c) optimal treatment assignment rules. An evaluation of multiple programmes of the Swiss Active Labour Market Policy illustrates how DML-based methods enable a comprehensive programme evaluation. Motivated by extreme individualised treatment effect estimates of the DR-learner, we propose the normalised DR-learner (NDR-learner) to address this issue. The NDR-learner acknowledges that individualised effect estimates can be stabilised by an individualised normalisation of inverse probability weights.

Keywords: *Causal machine learning, conditional average treatment effects, DR-learner, individualised treatment rules, multiple treatments, policy learning.*

JEL codes: C21.

1. INTRODUCTION

The adaptation of so-called machine learning to causal inference has been a productive area of methodological research in recent years. The resulting new methods complement the existing econometric toolbox for programme evaluation along at least two dimensions (see, for recent overviews, Athey and Imbens, 2017, 2019; Abadie and Cattaneo, 2018). On the one hand, they provide flexible methods to estimate standard average effects. In particular, they provide a data-driven approach to variable and model selection in studies that rely on an unconfoundedness assumption¹ for identification. On the other hand, they enable a more comprehensive evaluation by providing new methods for the flexible estimation of heterogeneous effects and of treatment assignment rules.

This paper considers double machine learning (DML) (Chernozhukov et al., 2018) as a framework for flexible and comprehensive programme evaluation. The DML framework seems attractive because (a) it can be combined with a variety of standard supervised machine learning methods; (b) it covers average effects for binary (e.g., Belloni et al., 2014; Belloni et al., 2017; Chernozhukov et al., 2018), multiple (e.g., Farrell, 2015) as well as continuous treatments (e.g., Kennedy et al., 2017; Colangelo and Lee, 2019; Semenova and Chernozhukov, 2021); (c) it

¹ Also known as exogeneity, selection on observables, ignorability, or conditional independence assumption.

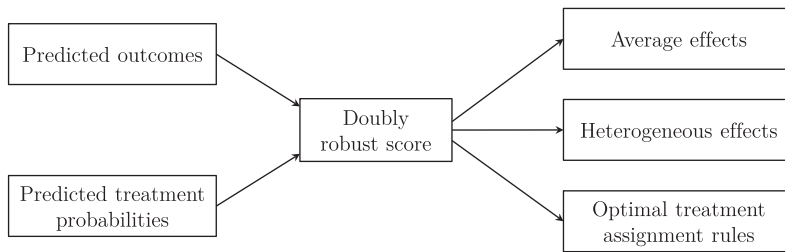


Figure 1. Stylised workflow of double machine learning-based programme evaluation.

naturally extends to the estimation of heterogeneous treatment effects of different forms like canonical subgroup effects, the best linear prediction of effect heterogeneity, or non-parametric effect heterogeneity (e.g., Fan et al., 2022; Zimmert and Lechner, 2019; Foster and Syrgkanis, 2019; Oprescu et al., 2019; Semenova and Chernozhukov, 2021; Kennedy, 2020; Curth and Van der Schaar, 2021); and (d) it can be used to estimate optimal treatment assignment rules (e.g., Dudik et al., 2011; Athey and Wager, 2021; Zhou et al., 2018). All these DML-based methods have favourable statistical properties and allow the use of standard tools like t -tests, OLS, kernel regression, series regression, or supervised machine learning for estimating causal parameters of interest after flexibly adjusting for confounding.

This paper starts with a review of DML-based methods, then applies these methods in a standard labour economic setting, and comes back to the methods by proposing the normalised DR-learner as a potential fix to a finite sample problem encountered in the application. Thus, it contributes to the steadily growing literature of causal machine learning for programme evaluation in three ways. First, the review highlights that methods for different parameters build on the same doubly robust score. The construction of this score might be computationally expensive because it requires the estimation of outcomes and treatment probabilities via machine learning methods. However, once constructed the score can be reused to estimate a variety of interesting parameters. This paper focuses on methods that build on the doubly robust score because they allow to leverage conceptual and computational synergies. The result is a comprehensive pipeline for programme evaluation within the same framework as Figure 1 illustrates. This is currently not possible with the variety of more specialised alternatives that integrate machine learning in the estimation of average treatment effects (e.g., Van der Laan and Rubin, 2006; Athey et al., 2018; Avagyan and Vansteelandt, 2017; Tan, 2020; Ning et al., 2020), heterogeneous treatment effects (e.g., Tian et al., 2014; Athey and Imbens, 2016; Chernozhukov et al., 2017; Wager and Athey, 2018; Athey et al., 2019; Künzel et al., 2019; Nie and Wager, 2021), and optimal treatment assignment (e.g., Bansak et al., 2018; Kallus, 2018).

Second, we use DML-based methods to provide a comprehensive and computationally convenient evaluation of four programmes of the Swiss Active Labour Market Policy (ALMP) in a standard dataset (Lechner et al., 2020). The evaluation in this paper illustrates the potential of DML-based methods for programme evaluations under unconfoundedness and provides a potential blueprint for similar analyses. This adds to a small but steadily growing literature that applies causal machine learning to programme evaluation in general (e.g., Bertrand et al., 2017; Strittmatter, 2018; Gulyas and Pytka, 2019; Knittel and Stolper, 2019; Davis and Heller, 2020; Baiardi and Naghi, 2021; Farbmacher et al., 2021), and to evaluations based on unconfoundedness

in particular (e.g., Kreif and DiazOrdaz, 2019; Cockx et al., 2020; Knaus et al., 2020; Knaus, 2021).

Third, we contribute to the methodological literature on the flexible estimation of individualised treatment effects (see, for a recent overview, Knaus et al., 2021) by proposing the normalised DR-learner (NDR-learner), which builds on the recent DR-learner of Kennedy (2020). The application reveals that the plain DR-learner produces few extreme effect estimates. It turns out that individualised effect estimates can be stabilised by an individualised normalisation of inverse probability weights. Thus, the NDR-learner can be considered as a generalisation of the popular Hájek (1971) normalisation for inverse probability weighting estimators for average effects. The increased stability comes at the price that the NDR-learner limits the class of permissible machine learning methods for effect heterogeneity estimation to methods that form predictions as convex combination of outcomes (e.g., random forests).

Overall, we find that DML-based methods provide a promising set of methods for programme evaluation. The estimated average programme effects are in line with the previous literature. We find that computer, vocational, and language courses increase employment in the 31 months after programmes start, while the effects of job search trainings are mostly negative. The heterogeneity analysis additionally reveals substantial heterogeneities by gender, nationality, previous labour market success, and qualification. These are picked up by the estimated optimal assignment rules.

The paper proceeds as follows. Section 2 defines the estimands of interest and their identification under unconfoundedness. Section 3 reviews DML-based methods for estimation and introduces the NDR-learner. Section 4 presents the application. Section 5 describes the implementation of the methods. Section 6 reports the results. Section 7 concludes. The Online Appendix provides additional explanations and results as well as an R replication notebook. The R-package *causal DML* implements the applied estimators.

2. ESTIMANDS OF INTEREST

2.1. Definition

We define the estimands of interest in the multiple treatment version of the potential outcomes framework (Rubin, 1974; Imbens, 2000; Lechner, 2001). Let $\mathcal{W} = \{0, \dots, T\}$ denote a set of multiple programmes and $D_i(w) = \mathbb{1}(W_i = w)$ a binary variable indicating in which programme individual i ($i = 1, \dots, N$) is actually observed.² We assume that each individual has a potential outcome $Y_i(w)$ for all $w \in \mathcal{W}$. Without loss of generality, the discussion below assumes that higher outcome values are desirable.

The first estimand of interest is the average potential outcome (APO), $\gamma_w = E[Y_i(w)]$. It answers the question about the average outcome if the whole population was assigned to programme w . However, the more interesting question is usually to compare different programmes w and w' . To this end, we take the difference of the according individual potential outcomes, $Y_i(w) - Y_i(w')$,³ and aggregate them to different estimands: first, the average treatment effect (ATE), $\delta_{w,w'} = E[Y_i(w) - Y_i(w')]$. Second, the average treatment effect on the treated (ATET), $\theta_{w,w'} = E[Y_i(w) - Y_i(w') \mid W_i = w]$. Third, the conditional average treatment effect (CATE),

² For DML-based estimation with continuous treatments see, e.g., Kennedy et al. (2017), Colangelo and Lee (2019), and Semenova and Chernozhukov (2021).

³ This would be $Y_i(1) - Y_i(0)$ in the canonical binary treatment setting.

$\tau_{w,w'}(z) = E[Y_i(w) - Y_i(w') \mid Z_i = z]$, where $Z_i \in \mathcal{Z}$ is a vector of observed pre-treatment variables.⁴

The different aggregations accommodate the notion that treatment effects might be heterogeneous. ATE represents the average effect in the population, while ATET shows it for the subpopulation that is actually observed in programme w . Thus, the comparison of ATE and ATET can be informative about the quality of the programme assignment mechanism. For example, ATET being larger than ATE indicates that the observed programme assignment is better than random.

The ATET is defined by the observed programme assignment and thus not subject to the choice of the researcher. In contrast, the conditioning variables Z_i of the CATE are specified by the researcher to investigate potentially heterogeneous effects across the groups of individuals that are defined by different values of Z_i . Such heterogeneous effects can be indicative for underlying mechanisms. Further, CATEs characterise which groups win and which lose by how much by receiving programme w instead of w' .

The different average effects above provide a comprehensive evaluation of programmes under the current programme assignment policy. In many applications, however, we want to conclude the analysis with a recommendation how the assignment policy could be improved. This can either be done using the evidence on the different average effects defined above or by formally defining the objective of an optimal assignment rule. The latter is pursued by the literature on statistical treatment rules (e.g., Manski, 2004; Hirano and Porter, 2009; Stoye, 2009, 2012; Kitagawa and Tetenov, 2018; Athey and Wager, 2021 and references therein). Here we focus on the case with multiple treatment options as considered by Zhou et al. (2018).

Let $\pi(Z_i)$ be a policy that assigns individuals to programmes according to their characteristics Z_i or, put more formally, the function $\pi(Z_i)$ maps observable characteristics to a programme: $\pi : \mathcal{Z} \rightarrow \mathcal{W}$. In principle, the policy rule can be completely flexible and in the ideal world we would assign each individual to the programme with the highest conditional APO, $E[Y_i(w) \mid Z_i = z]$. However, in many cases we want to restrict the set of candidate policy rules denoted by Π to be interpretable for the communication with decision makers or to incorporate costs or fairness constraints. Each of these candidate policy rules has a policy value function denoted by $Q(\pi) = E[Y_i(\pi(Z_i))] = E[\sum_w \mathbb{1}(\pi(Z_i) = w)Y_i(w)]$. $Q(\pi)$ quantifies the average population outcome if policy rule π would be used to assign programmes. The estimand of interest is then the optimal policy rule π^* with the highest value function for the set of candidate policy rules, or formally $\pi^* = \arg \max_{\pi \in \Pi} Q(\pi)$.

2.2. Identification

The previous section defined the estimands of interest in terms of potential outcomes. However, each individual is only observed in one programme. Thus, only one potential outcome per individual is observable and the other potential outcomes remain latent. This is the fundamental problem of causal inference (Holland, 1986) and we need further assumptions to identify the estimands of interest. In this paper, we consider the unconfoundedness assumption that assumes access to a vector of pre-treatment variables $X_i \in \mathcal{X}$ containing Z_i such that the following standard assumptions hold (e.g., Imbens and Rubin, 2015):

⁴ We focus in this study on expectations of the individual treatment effects. DML-based methods for quantile treatment effects can be found, e.g., in Belloni et al. (2017) and Kallus et al. (2019).

ASSUMPTION 2.1.

- (a) *Unconfoundedness*: $Y_i(w) \perp\!\!\!\perp W_i \mid X_i = x, \forall w \in \mathcal{W}, \text{ and } x \in \mathcal{X}$.
 (b) *Common support*: $0 < P[W_i = w \mid X_i = x] \equiv e_w(x), \forall w \in \mathcal{W} \text{ and } x \in \mathcal{X}$.
 (c) *Stable Unit Treatment Value Assumption (SUTVA)*: $Y_i = Y_i(W_i)$.

The unconfoundedness assumption requires that X_i contains all confounding variables that jointly affect programme assignment and the outcome. Common support states that it must be possible to observe each individual in all programmes. SUTVA rules out interference. These assumptions allow the identification of the average potential outcome (APO) conditional on confounders in three common ways:

$$E[Y_i(w) \mid X_i = x] = E[Y_i \mid W_i = w, X_i = x] \equiv \mu(w, x) \quad (2.1)$$

$$= E \left[\frac{D_i(w)Y_i}{e_w(x)} \mid X_i = x \right] \quad (2.2)$$

$$= E \left[\underbrace{\mu(w, x) + \frac{D_i(w)(Y_i - \mu(w, x))}{e_w(x)}}_{\equiv \Gamma(w, x)} \mid X_i = x \right]. \quad (2.3)$$

Equation (2.1) shows that the conditional APO is identified as a conditional expectation of the observed outcome. Equation (2.2) shows that it is identified by reweighting the observed outcome with the inverse treatment probability. Finally, equation (2.3) adds the reweighted outcome residual to the conditional outcome representation of equation (2.1). This seems redundant because we can check that the reweighted residual has expectation zero under unconfoundedness. However, this identification result is doubly robust in the sense that it still holds if we replace either $\mu(w, x)$ or $e_w(x)$ in equation (2.3) by arbitrary functions of x .⁵ This doubly robust structure plays a crucial role for the estimation procedures that we discuss in the next section.

From an identification perspective, $\Gamma(w, x)$ defined in equation (2.3) suffices to identify all estimands of interest stated in the previous subsection:

- (a) APO: $\gamma_w = E[Y_i(w)] = E[\Gamma(w, X_i)]$
 (b) ATE: $\delta_{w, w'} = E[Y_i(w) - Y_i(w')] = E[\Gamma(w, X_i) - \Gamma(w', X_i)]$
 (c) ATET: $\theta_{w, w'} = E[Y_i(w) - Y_i(w') \mid W_i = w] = E[\Gamma(w, X_i) - \Gamma(w', X_i) \mid W_i = w]$
 (d) CATE: $\tau_{w, w'}(z) = E[Y_i(w) - Y_i(w') \mid Z_i = z] = E[\Gamma(w, X_i) - \Gamma(w', X_i) \mid Z_i = z]$
 (e) Policy value: $Q(\pi) = E[Y_i(\pi(Z_i))] = E[\sum_w \mathbb{1}(\pi(Z_i) = w)\Gamma(w, X_i)]$
 (f) Optimal policy: $\pi^* = \arg \max_{\pi \in \Pi} Q(\pi) = \arg \max_{\pi \in \Pi} E[\sum_w \mathbb{1}(\pi(Z_i) = w)\Gamma(w, X_i)]$

3. ESTIMATION BASED ON DOUBLE MACHINE LEARNING

3.1. The doubly robust scores

All double machine learning-based estimators for the estimands of interest build on the doubly robust scores of Robins et al. (1994, 1995) and their Augmented Inverse Probability Weighting (AIPW) estimator in particular. In the following, large Greek letters denote the scores corresponding to the small Greek letters used to define the estimands in Section 2.1.

⁵ Online Appendix S1 reviews identification and identification double robustness of equation (2.3) for completeness.

The construction of the doubly robust scores requires the input of so-called nuisance parameters that are usually of secondary interest and considered as tool to eventually obtain the parameters of interest. In our case, the two nuisance parameters are $\mu(w, x) = E[Y_i | W_i = w, X_i = x]$ and $e_w(x) = P[W_i = w | X_i = x]$ for all w . $\mu(w, x)$ is the conditional outcome mean for the subgroup observed in programme w . $e_w(x)$ is the conditional probability to be observed in programme w , also known as the propensity score. Usually these functions are unknown and need to be estimated. Following Chernozhukov et al. (2018) they are estimated based on K -fold cross-fitting: (a) randomly divide the sample in K folds of similar size; (b) leave out fold k and estimate models for the nuisance parameters in the remaining $K - 1$ folds; (c) use these models to predict $\hat{\mu}^{-k}(w, x)$ and $\hat{e}_w^{-k}(x)$ in the left out fold k ; and (d) repeat (a) to (c) such that each fold is left out once. This procedure avoids overfitting in the sense that no observation is used to predict its own nuisance parameters. To avoid notational clutter, we ignore the dependence on the specific fold in the following notation and refer to the cross-fitted nuisance parameters as $\hat{\mu}(w, x)$ and $\hat{e}_w(x)$.

The main building block of the following estimators is the doubly robust score of the APO, which replaces the true nuisance parameters in equation (2.3) by their cross-fitted predictions:

$$\hat{\Gamma}_{i,w} = \hat{\mu}(w, X_i) + \frac{D_i(w)(Y_i - \hat{\mu}(w, X_i))}{\hat{e}_w(X_i)}. \quad (3.1)$$

The ATE score for the comparison of treatment w and w' is then constructed as the difference of the respective APO scores:

$$\hat{\Delta}_{i,w,w'} = \hat{\Gamma}_{i,w} - \hat{\Gamma}_{i,w'} \quad (3.2)$$

The only estimator we consider that uses the same nuisance parameter but plugs them into a different score is the ATET estimator. Although the identification result with the doubly robust APO score in the previous section holds, it is not doubly robust. However, the doubly robust score for the ATET exists and is defined as

$$\hat{\Theta}_{i,w,w'} = \frac{D_i(w)(Y_i - \hat{\mu}(w', X_i))}{\hat{e}_w} - \frac{D_i(w')\hat{e}_w(X_i)(Y_i - \hat{\mu}(w', X_i))}{\hat{e}_w\hat{e}_{w'}(X_i)}, \quad (3.3)$$

where $\hat{e}_w = N_w/N$ is the unconditional treatment probability with N_w counting the number of individuals observed in programme w (see also, e.g., Farrell, 2015).

3.2. Average potential outcomes and treatment effects

The estimation of the APOs, ATEs, and ATETs boils down to taking the means of the previously defined doubly robust scores. For statistical inference, we can rely on standard one-sample t -tests. Thus, the score's mean and the variance of this mean are the point and the variance estimate of the respective estimand of interest:

- (a) APO: $\hat{\mu}_w = N^{-1} \sum_i \hat{\Gamma}_{i,w}$ and $\hat{\sigma}_{\mu_w}^2 = N^{-1} \sum_i (\hat{\Gamma}_{i,w} - \hat{\mu}_w)^2$
- (b) ATE: $\hat{\delta}_{w,w'} = N^{-1} \sum_i \hat{\Delta}_{i,w,w'}$ and $\hat{\sigma}_{\delta_{w,w'}}^2 = N^{-1} \sum_i (\hat{\Delta}_{i,w,w'} - \hat{\delta}_{w,w'})^2$
- (c) ATET: $\hat{\theta}_{w,w'} = N^{-1} \sum_i \hat{\Theta}_{i,w,w'}$ and $\hat{\sigma}_{\theta_{w,w'}}^2 = N^{-1} \sum_i (\hat{\Theta}_{i,w,w'} - \hat{\theta}_{w,w'})^2$

Note that the estimated variances require no adjustment for the fact that we have estimated the nuisance parameters in a first step. The resulting estimators are consistent, asymptotically normal, and semiparametrically efficient under the main assumption that the estimators of the cross-fitted nuisance parameters are consistent and converge sufficiently fast (Belloni et al., 2014; Farrell, 2015; Belloni et al., 2017; Chernozhukov et al., 2018). In particular, the product of the convergence rates of the outcome and propensity score estimators must be faster than $n^{1/2}$. This allows to apply machine learning to estimate the nuisance parameters.⁶ Flexible machine learning estimators converge usually slower than the parametric rate $n^{1/2}$ but several are known to be able to achieve $n^{1/4}$ and faster, which would be sufficient if both nuisance parameter estimators achieve it.⁷

It is well known that estimators using doubly robust scores and parametric models for the nuisance parameters are doubly robust in the sense that they remain consistent if one of the parametric models is misspecified (see, e.g., Glynn and Quinn, 2009). The difference of the DML version is that it exploits what Smucler et al. (2019) call ‘rate double robustness’. This robustness allows to estimate the parameters of interest at the parametric rate $n^{1/2}$ even if the nuisance parameters are estimated at slower rates using machine learning methods that do not require the specification of an actual parametric model.

The rate double robustness is the consequence of the so-called Neyman orthogonality of the doubly robust score. Neyman orthogonality is at the heart of the general DML framework of Chernozhukov et al. (2018). Scores with this orthogonality are immune against small errors in the estimation of nuisance parameters and thus allow them to be estimated via machine learning. Online Appendix S1.2 revisits what this means in formal terms.

3.3. Conditional average treatment effects

3.3.1. DR-learner. We can reuse the ATE score of equation (3.2) to estimate conditional effects. The so-called DR-learner was introduced for binary treatments but directly translates also to multiple treatments settings. It exploits that the conditional expectation of the score with known nuisance parameters equals CATE: $\tau_{w,w'}(z) = E[\Delta_{i,w,w'} \mid Z_i = z]$.⁸ Thus, a natural way to estimate CATEs is to use the score with estimated nuisance parameters, $\hat{\Delta}_{i,w,w'}$, as pseudo-outcome in a general regression framework:

$$\hat{\tau}^{dr}(Z_i) = \arg \min_{\tau} \sum_{i=1}^N (\hat{\Delta}_{i,w,w'} - \tau(Z_i))^2. \quad (3.4)$$

From a conceptual and estimation perspective it is instructive to distinguish two special cases of CATEs at this point (see also Knaus et al., 2021): (a) Group average treatment effects (GATE) provide the average effects for pre-specified, usually low-dimensional, groups.⁹ This covers standard subgroup analysis comparing, e.g., effects of men and women, or heterogeneity along

⁶ Further results, regularity conditions and discussions can be found in Chernozhukov et al. (2018, sect. 5.1).

⁷ For example, versions of Lasso (Belloni and Chernozhukov, 2013), Boosting (Luo and Spindler, 2016), Random Forests (Wager and Walther, 2015; Syrgkanis and Zampetakis, 2020), Neural Nets (Farrell et al., 2021), forward model selection (Kozbur, 2020), or ensembles of those can be shown to achieve the required rates under conditions stated in the original papers.

⁸ Note that this does not work for the ATET score in equation (3.3) and suitable adaptations are beyond the scope of this paper.

⁹ Note that the GATE is different to the sorted group average treatment effect (GATES) of Chernozhukov et al. (2017).

pre-specified continuous variables like age. (b) Individualised average treatment effects (IATEs) aim for the most detailed effect heterogeneity considering all confounders as heterogeneity variables, i.e., $Z_i = X_i$ and thus $IATE(x) = \tau_{w,w'}(x) = E[Y_i(w) - Y_i(w') \mid X_i = x]$.

OLS, series, or kernel regressions of the pseudo-outcome on low-dimensional heterogeneity variables estimate *GATEs*. The outputs of such regressions can be interpreted in the standard way. The only difference is that instead of modelling the level of an outcome, they now model the level of a causal effect. Most importantly standard statistical inference applies as is shown for OLS and series regression by Semenova and Chernozhukov (2021) as well as for kernel regression by Fan et al. (2022) and Zimmert and Lechner (2019). Similar to the discussion in the previous section, the Neyman orthogonality of $\Delta_{i,w,w'}$ allows to ignore that nuisance parameters are estimated with flexible methods potentially converging slower than $n^{1/2}$ when calculating standard errors. The details about the required convergence rates are discussed in the referenced papers.

IATEs may be estimated using the pseudo-outcome in supervised machine learning regressions with the full set of confounders as predictors. As discussed by Chernozhukov et al. (2017) statistical inference is not yet well understood for low-dimensional Z_i and even harder for high-dimensional Z_i when machine learning is used to solve equation (3.4). However, Kennedy (2020) shows that the doubly robust structure of the ATE score results in favourable bounds on the mean squared error for the estimated IATEs that would not be attainable by outcome regression or IPW-based methods alone.¹⁰

We consider two variants of the DR-learner for *IATEs*. First, we reuse the pseudo-outcome in one supervised machine learning regression to estimate IATEs in-sample. This *full sample procedure* is computationally convenient but prone to overfitting. Thus, the second variant produces out-of-sample IATE predictions for each individual in the sample. Following Algorithm 1 of Kennedy (2020), this requires a *four-fold cross-fitting* scheme that is detailed in Algorithm S2.1 of Online Appendix S2. The computational downside of this procedure is that we cannot reuse the same nuisance parameter predictions as for the average estimator and need to estimate them for the IATE only. However, the results below suggest that this computational effort is important to avoid severe overfitting.

3.3.2. Normalised DR-learner. Note that the point estimates of the plain DR-learner can be expressed as $\hat{\tau}^{dr}(z) = \sum_{i=1}^N \alpha_i \hat{\Delta}_{i,w,w'}$ if the weight α_i that each observation receives can be calculated. For example, the ATE estimator as special case of the DR-learner with Z_i being a constant uses $\alpha_i = 1/N$, the least squares regression uses $\alpha_i = z(\mathbf{Z}'\mathbf{Z})^{-1}\mathbf{Z}'$ with $\mathbf{Z}$ being the stacked covariate matrix, and the kernel regression uses $\alpha_i = \frac{\mathcal{K}_h(Z_i - z)}{\sum_{i=1}^N \mathcal{K}_h(Z_i - z)}$ with $\mathcal{K}_h$ representing a proper kernel function. The class of estimators with a known weighted representation is called linear smoothers (see e.g., Buja et al., 1989). Popular machine learners like tree-based methods (regression trees, random forests, or boosted trees), Ridge or any method that runs OLS after variable selection like Post-Lasso (Belloni and Chernozhukov, 2013) have this structure.¹¹ Also for these methods we know the weight $\alpha_i(x)$ that each observation receives in predicting the (pseudo-)outcome at x . These weights usually sum up to one, i.e., $\sum_{i=1}^N \alpha_i(x) = 1$. Using such

¹⁰ The Orthogonal Random Forest of Oprescu et al. (2019) is another estimator that is based on the pseudo-outcome idea and can be asymptotically normal under the assumption of parametric nuisance parameters. We focus in this paper on the more general DR-learner. See also Curth and Van der Schaar (2021) for a more nuanced analysis of the DR-learner in comparison to other alternatives.

¹¹ In practice most of these methods are applied with data-driven selection of tuning parameters, which makes them strictly speaking non-linear smoothers (Buja et al., 1989). However, this does not affect our results.

outcome weighting predictors in the final step allows to express the DR-learner estimated IATE as

$$\begin{aligned}\hat{\tau}_{w,w'}^{drl}(x) &= \sum_{i=1}^N \alpha_i(x) \hat{\Delta}_{i,w,w'} \\ &= \sum_{i=1}^N \alpha_i(x) [\hat{\mu}(w, X_i) - \hat{\mu}(w', X_i)] \\ &\quad + \sum_{i=1}^N \underbrace{\frac{\alpha_i(x) D_i(w)}{\hat{e}_w(X_i)}}_{\lambda_i^w(x)} \tilde{Y}_i(w, X_i) - \sum_{i=1}^N \underbrace{\frac{\alpha_i(x) D_i(w')}{\hat{e}_{w'}(X_i)}}_{\lambda_i^{w'}(x)} \tilde{Y}_i(w', X_i),\end{aligned}\quad (3.5)$$

where $\tilde{Y}_i(w, X_i) = Y_i - \hat{\mu}(w, X_i)$ denotes the outcome residual.

The DR-learner shares the problem of all estimators that involve reweighting by the inverse of the propensity score. In finite samples, $\lambda_i^w(x)$ and $\lambda_i^{w'}(x)$ usually do not sum to one, i.e., $\sum_{i=1}^N \lambda_i^w(x) \neq 1$ and $\sum_{i=1}^N \lambda_i^{w'}(x) \neq 1$. This is especially problematic if it sums to something much greater than one. In this case the weighted residuals receive much more weight than the outcome regressions. This might result in implausibly large effect estimates that even could fall outside of the possible bounds of a given outcome variable (Kang and Schafer, 2007; Robins et al., 2007).¹²

For average effects the Hájek (1971) normalisation is recommended to stabilise estimators using inverse probability weights (e.g., Imbens, 2004; Lunceford and Davidian, 2004; Robins et al., 2007; Busso et al., 2014). However, equation (3.5) highlights that the inverse probability weights become x -specific and such a one time normalisation that targets the average effect does not solve the problem for the individualised effect. This can be problematic as finite sample imbalances are more likely to occur on the individualised level. Thus, we propose the normalised DR-learner (NDR-learner) as a stabilised complement to the DR-learner.

The *NDR-learner* normalises the weighted residuals by the sum of weights:

$$\begin{aligned}\hat{\tau}_{w,w'}^{ndrl}(x) &= \sum_{i=1}^N \alpha_i(x) [\hat{\mu}(w, X_i) - \hat{\mu}(w', X_i)] \\ &\quad + \left(\sum_{i=1}^N \lambda_i^w(x) \right)^{-1} \sum_{i=1}^N \lambda_i^w(x) \tilde{Y}_i(w, X_i) \\ &\quad - \left(\sum_{i=1}^N \lambda_i^{w'}(x) \right)^{-1} \sum_{i=1}^N \lambda_i^{w'}(x) \tilde{Y}_i(w', X_i).\end{aligned}\quad (3.6)$$

This ensures that the weights of the residuals sum up to one under the condition that weights $\alpha_i(x)$ are non-negative. Thus, methods like Ridge or Post-Lasso with potentially negative weights might not be applicable.

The NDR-learner is more demanding from a computational point of view because it requires to calculate the weights $\alpha_i(x)$ and the normalisation for each x of interest (Algorithm S2.2 in Online

¹² For bounded outcomes, the effects must lie in the interval $[Y_{\min} - Y_{\max}, Y_{\max} - Y_{\min}]$, with $Y_{\min}$ and $Y_{\max}$ denoting the minimum and maximum values of the outcome, respectively.

Table 1. Example of DML-based optimal treatment assignment.

i	Z_i	π^1	π^2	π^3	π^4	$\hat{Q}(\pi^1)$	$\hat{Q}(\pi^2)$	$\hat{Q}(\pi^3)$	$\hat{Q}(\pi^4)$
1	0	0	0	1	1	$\hat{\Gamma}_{1,0}$	$\hat{\Gamma}_{1,0}$	$\hat{\Gamma}_{1,1}$	$\hat{\Gamma}_{1,1}$
2	1	0	1	0	1	$\hat{\Gamma}_{2,0}$	$\hat{\Gamma}_{2,1}$	$\hat{\Gamma}_{2,0}$	$\hat{\Gamma}_{2,1}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Appendix S2 provides the details of the implementation). However, the application below shows that the normalisation deals well with the cases where outcome residuals receive high weights leading to implausibly large effect estimates. Thus, the NDR-learner is an interesting alternative to the DR-learner if effect sizes become suspicious.

3.4. Optimal treatment assignment

The APO score of Section 3.1 can also be reused to estimate optimal treatment assignment. To this end, note that the value function of any policy rule $\pi(Z_i)$ can be estimated as

$$\hat{Q}(\pi) = N^{-1} \sum_{i=1}^N \sum_{w=0}^T \mathbb{1}(\pi(Z_i) = w) \hat{\Gamma}_{i,w}.$$

This means each individual contributes the score of the treatment that they are assigned to under this policy rule. However, we are not necessarily interested in the value function of some policy rule, but want to estimate the optimal policy rule that maximises this value function, $\hat{\pi}^* = \arg \max_{\pi \in \Pi} \hat{Q}(\pi)$. This requires to search over all candidate policy rules to find the optimum as there exists no closed form solution.

Example: Consider the case where Z_i is a binary covariate and W_i is a binary treatment. We have four different policy rules: treat nobody (π^1), treat only those with $Z_i = 1$ (π^2), treat only those with $Z_i = 0$ (π^3), or treat everybody (π^4). We illustrate this using two representative observations, $i = 1$ with $Z_1 = 0$, and $i = 2$ with $Z_2 = 1$ in Table 1. The columns three to six show the assignments under the four potential assignment rules. For example, the first observation receives no treatment under policy rules π^1 and π^2 , but is treated under policy rules π^3 and π^4 . To find the optimal rule, we compare the means of the APO scores in the last four columns and pick the policy rule that corresponds to the largest mean. The number of policy values to compare increases dramatically in settings with multiple treatments and Z_i being a vector of potentially non-binary variables.

We expect that the estimated policy in finite samples and with estimated nuisance parameters does not coincide with the true optimal policy rule. This is conceptualised as the ‘regret’ defined as the difference between the true and the estimated optimal value function, $R(\hat{\pi}^*) = Q(\pi^*) - Q(\hat{\pi}^*)$.

Zhou et al. (2018) show that the DML-based procedure minimises the maximum regret asymptotically under two main conditions: first, the same convergence conditions for the nuisance parameters that are required for ATE estimation (the product of the nuisance parameter convergence rates achieves $n^{1/2}$). Second, the set of candidate policy rules Π is not too complex. In particular, Zhou et al. (2018) show that decision trees with fixed depth are a suitable class of policy rules. Again the double robustness of the used scores results in statistical guarantees that are not achievable for methods based on outcome regressions or IPW alone.

Table 2. Descriptive statistics of selected variables by programme type.

	No programme (1)	Job search (2)	Vocational (3)	Computer (4)	Language (5)
No. of observations	47,620	11,610	858	905	1,504
Outcome: months employed of 31	14.7	14.4	18.4	19.2	13.5
Female (binary)	0.44	0.44	0.33	0.60	0.55
Age	36.6	37.3	37.5	39.1	35.3
Foreigner (binary)	0.36	0.33	0.30	0.21	0.66
Employability	1.93	1.98	1.93	1.97	1.85
Past income in CHF10,000	4.25	4.67	4.87	4.32	3.73

Notes: Employability is an ordered variable with one indicating low employability, two medium employability, and three high employability. The exchange rate USD/CHF was roughly 1.3 at that time. The full set of variables is reported in Table S1 of Online Appendix S3.

4. APPLICATION: SWISS ACTIVE LABOUR MARKET POLICY

We use a standard observational dataset of Swiss Active Labour Market Policy (ALMP) that is already the basis of previous studies (Huber et al., 2017; Lechner, 2018; Knaus et al., 2020) to estimate the effect of different programmes on employment.¹³ In particular, we start with the sample of 100,120 unemployed individuals of Huber et al. (2017) that consists of 24 to 55 year old individuals registered unemployed in 2003.¹⁴ We consider non-participants and participants of four different *programme types*: job search, vocational training, computer programmes, and language courses.¹⁵ As the assignment policies differ substantially across the three language regions, we focus only on individuals living in the German-speaking part and remove those in the French- and Italian-speaking part to avoid common support problems.

We evaluate the first programme participation within the first six months after the begin of the unemployment spell. One problem of this definition is that non-participants comprise people that quickly come back into employment before they would be assigned to a training programme. This could result in an overly optimistic evaluation of non-participation. We follow Lechner (1999) and Lechner and Smith (2007) and assign pseudo programme starting points to the non-participants and keep only those who are still unemployed at this point.¹⁶ This results in a final sample size of 62,497 observations.

The *outcome* of interest is the cumulated number of months in employment in the 31 months after programme start, which is the maximum available time span in the dataset. Row one of Table 2 provides the number of observations in each group. Roughly 75% participate in no programme. By far the largest programme is the job search programme, which is also called a basic programme. The more specific programmes are much smaller with roughly 1,000 observations each. Row two shows that the average outcomes substantially differ by different groups. However,

¹³ Gerfin and Lechner (2002), Lalive et al. (2008), and Knaus et al. (2020), among others, provide a more detailed description of the surrounding institutional setting.

¹⁴ The dataset is available as restricted use file via the platform FORSbase (Lechner et al., 2020).

¹⁵ The dataset contains also participants of an employment programme and personality training. However, we leave them out to keep the number of obtained results manageable.

¹⁶ The assignment of the pseudo starting point is based on estimated probabilities to start a programme at a specific time. The probability depends also on covariates and is estimated using the same random forest specification that is discussed later in Section 5.

it is not clear whether this is only due to selection effects because the observable characteristics are not comparable across groups, as the remaining rows show. The share of females, the share of foreigners, and past income, especially, differ quite substantially across programmes. The *confounders* comprise 45 variables and are reported in Table S1 of Online Appendix S3. They consist of socio-economic characteristics of the unemployed individuals, caseworker characteristics, information about the assignment process, information about the previous job, and regional economic indicators.

5. IMPLEMENTATION

The nuisance parameters are estimated via random forest (Breiman, 2001) using the implementation with honest splitting in the *grf* R-package (Athey et al., 2019) and five-fold cross-fitting. The tuning parameters in each regression are selected by out-of-bag validation. All regressions apply the full set of confounders. We run the outcome regressions for each treatment group separately to obtain $\hat{\mu}(w, x)$. Also the propensity scores are separately estimated for each treatment using a treatment indicator as outcome in the random forest. The propensity scores are then normalised to sum to one within an individual.

We estimate CATEs at different granularity. First, we investigate GATEs for subgroups by gender, foreigners, and three categories of employability. These are regularly used in the programme evaluation literature and usually investigated by re-estimating everything in the subgroups. However, it can be performed at very low computational costs after DML for average effects using only a standard OLS regression with the pseudo-outcome as described in Section 3.3.1 and using dummy variables for all groups but the reference group as covariates. Second, we estimate kernel regression and spline regression GATEs for the continuous variables age and past income based on the R-packages *np* (Hayfield and Racine, 2008) and *crs* (Racine and Nie, 2021), respectively. The kernel regressions apply a second-order Gaussian kernel function and use 0.9 of the cross-validated bandwidth for undersmoothing as suggested by Zimmert and Lechner (2019). The spline regressions use B-splines with cross-validated degree and number of knots. Third, we specify an OLS regression in which all the five previously used variables enter linearly. Finally, we go beyond the handpicked variables and estimate the IATEs using all 45 confounders in the DR-learner and the NDR-learner. Both are implemented with the honest Random Forest because the *grf* package allows to extract the prediction weights $\alpha_i(x)$ required for the NDR-learner. We apply both variants described in Section 3.3.1. Once we estimate the IATE for each observation using DR- and NDR-learner in the full sample and once we predict them out-of-sample. For the latter, Online Appendix S2 provides a detailed description of the underlying DR- and NDR-learner algorithms.

The optimal treatment assignment rule is estimated as decision trees of depth one, two, and three. We follow Algorithm 2 for an exact tree search of Zhou et al. (2018) that is implemented in the *policytree* R-package (Sverdrup et al., 2020). We estimate the trees first with the five handpicked variables. However, these variables include gender and foreigner status that might be too sensitive to include in practice. Thus, we investigate another set of 16 variables that includes only the objective measures of education and labour market history of the unemployed persons that would be available for recommendations from the administrative records.

Table 3 summarises all required implementation steps. It highlights that a comprehensive DML-based programme evaluation can be run with few lines of code in any statistical software program that is capable of the operations in the third column. Thus, researchers can build their customised

Table 3. Steps of implementation.

Step	Input	Operation	Output
1.	W_i, X_i	Predict treatment probabilities	$\hat{e}_w(x)$
2.	Y_i, W_i, X_i	Predict treatment specific outcomes	$\hat{\mu}(w, x)$
3.	$Y_i, W_i, \hat{e}_w(x), \hat{\mu}(w, x)$	Plug into equation (3.1)	$\hat{\Gamma}_{i,w}$
4.	$\hat{\Gamma}_{i,w}$	Mean, one-sample t -test	APOs
5.	$\hat{\Gamma}_{i,w}$	Take difference	$\hat{\Delta}_{i,w,w'}$
6.	$\hat{\Delta}_{i,w,w'}$	Mean, one-sample t -test	ATEs
7.	$\hat{\Delta}_{i,w,w'}, Z_i$	OLS/Kernel/Series regression	GATEs
8.	$\hat{\Delta}_{i,w,w'}, X_i$	Supervised machine learning	IATEs
9.	$\hat{\Gamma}_{i,w}, Z_i$	Optimal decision tree	Optimal treatment rule

analyses in a modular fashion based on established code. Alternatively, the R-package causalDML already implements the required steps as showcased in the replication notebook accompanying this paper. Most importantly, the package provides a fast implementation of the individualised normalisation required for the NDR-learner in C++.

6. RESULTS

6.1. Average effects

We focus here on the effect estimates and discuss the nuisance parameters in Online Appendix S3.2. Throughout this section, we compare the four programmes to non-participation.¹⁷ Recall that the outcome of interest is the cumulated number of months employed in the 31 months after programme start. Figure 2 depicts ATE and ATET estimates and shows substantial differences in the effectiveness of programmes. The job search programme decreases the months in employment on average by about one month. In contrast, other programmes that teach hard skills show substantial improvements with roughly three additional months in employment on average.¹⁸

Comparing ATE and ATET shows no big differences for most programmes. This suggests that there is either no effect heterogeneity correlated with observables or that the assignment does not take advantage of this heterogeneity. We would expect to see ATETs being higher than ATEs if programme assignment is well targeted. However, we find only evidence for the opposite as the actual participants of a language course show a 1.5 months lower treatment effect compared to the population. This difference suggests that there is substantial effect heterogeneity to uncover and the potential to improve treatment assignment.

6.2. Heterogeneous effects

6.2.1. Subgroup GATEs. This subsection studies effect heterogeneity at different granularity. We start by estimating group average treatment effects (GATEs) for discrete subgroups. Panel A of Table 4 shows the result of an OLS regression with a female dummy as covariate, $\hat{\Delta}_{i,w,w'} =$

¹⁷ The underlying APOs are shown in Figure S2 of Online Appendix S3.3.
¹⁸ For a better understanding of the underlying dynamics, Figure S3 of Online Appendix S3.3 reports and discusses the effects of programme participation on the employment probabilities over time.

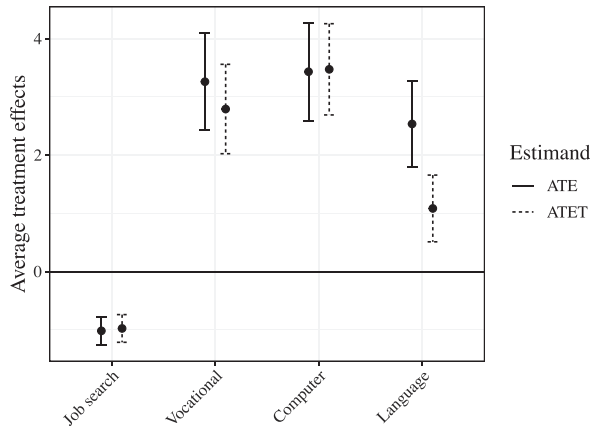


Figure 2. Average treatment effects of participation versus non-participation. The figure shows the point estimates of the average treatment effects of participating in the programme labelled on the x-axis versus non-participation and their 95% confidence intervals. Numeric results in Panels B and C of Table S5 of Online Appendix S3.3.

Table 4. Group average treatment effects.

	Job search (1)	Vocational (2)	Computer (3)	Language (4)
<i>Panel A: Female</i>				
Constant	− 1.29*** (0.17)	3.82*** (0.55)	2.33*** (0.60)	3.40*** (0.46)
Female	0.60** (0.25)	− 1.27 (0.87)	2.49*** (0.85)	− 1.97** (0.77)
<i>Panel B: Foreigner</i>				
Constant	− 1.27*** (0.16)	2.48*** (0.53)	3.75*** (0.50)	3.56*** (0.52)
Foreigner	0.70*** (0.26)	2.17** (0.90)	− 0.88 (0.93)	− 2.84*** (0.72)
<i>Panel C: Employability</i>				
Constant	− 0.18 (0.33)	5.48*** (1.04)	5.76*** (1.10)	2.61*** (0.85)
Medium employability	− 0.93*** (0.36)	− 2.41** (1.16)	− 2.67** (1.20)	− 0.18 (0.96)
High employability	− 1.50*** (0.50)	− 4.42*** (1.52)	− 3.59** (1.69)	0.59 (1.48)
F-statistic	5.04***	4.31**	2.96*	0.18

Notes: This table shows OLS coefficients and their heteroscedasticity robust standard errors (in parentheses) of regressions run with the pseudo-outcome defined as described in Section 3.3. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

$\beta_0 + \beta_1 \text{female}_i + \text{error}_i$. The constant (β_0) provides the GATE for the reference group men and the female coefficient (β_1) describes how much the GATE differs for women. The results show substantial gender differences in the effectiveness of programmes. Women significantly suffer less or profit more from job search and computer programme participation. This gender gap in the effectiveness of ALMPs is also well-documented in the literature (Crépon and Van den Berg, 2016; Card et al., 2018). In contrast to this, we find that women profit on average significantly less from language courses than men.

Panel B replaces the female dummy in the regression by a foreigner dummy. Strikingly, Swiss citizens as reference group show a big positive effect for participating in language courses but the effect disappears for foreigners. After adding the coefficient for foreigners to the constant, the foreigners' GATE is only 0.72 ($3.56 - 2.84$, standard error: 0.69). Crucial information to better understand this finding would be to know which languages they learn.¹⁹ However, this information is unfortunately not available in the dataset.

Panel C shows the results of a similar regression, but now with two dummies indicating medium and high employability such that low employability becomes the reference group. The F -statistic in the last line tests the joint significance of the two dummies. It is statistically significant at least at the 10% level for the programmes in the first three columns. They all show a common gradient that individuals with low employability benefit substantially more or at least suffer less from programme participation.

6.2.2. Nonparametric GATEs for continuous heterogeneity variables. While subgroup analyses are standard in programme evaluations, the estimation of non-parametric GATEs using kernel or series regression is rarely pursued. We estimate such GATEs along two continuous variables past income and age. We find no notable heterogeneity for the latter.²⁰ However, effect sizes are clearly associated with past income. Figure 3 shows on the left the results of the kernel regression and on the right of the series regression. Both estimators agree by and large. They document that effects decrease with higher past income for all but for language programmes. The latter have only a small positive effect for individuals with low past income but it increases with higher income. One potential explanation for these findings is that the value of language skills is larger for high-skilled workers in multilingual countries like Switzerland because they reduce information costs across language borders (see, e.g., Ipsphording, 2014).

6.2.3. Best linear prediction of GATEs. The GATEs considered so far were non-parametric, but only univariate. Now we model the GATE by specifying a multivariate OLS regression with the previously used covariates entering linearly. It is most likely misspecified and thus estimates the best linear predictor (BLP) of GATEs with respect to these variables. However, it provides a compact and accessible summary of the effect heterogeneities. Additionally, it holds the other included variables constant. Consider for example the coefficients for being female in Table 5. Compared to Table 4, the coefficients in the first three columns are smaller and the one for language courses is larger (for example, for job search it is 0.25 instead of 0.60). The reason is that it represents a partial effect that holds other variables like past income fixed. The subgroup female coefficient in Table 4 partly picks up that women have lower past income and that lower income is associated with higher treatment effects for all but language courses. This example

¹⁹ See Heiler and Knaus (2021) for a discussion about how treatment heterogeneity could drive effect heterogeneity.

²⁰ Figure S4 of Online Appendix S3.4 shows the according results.

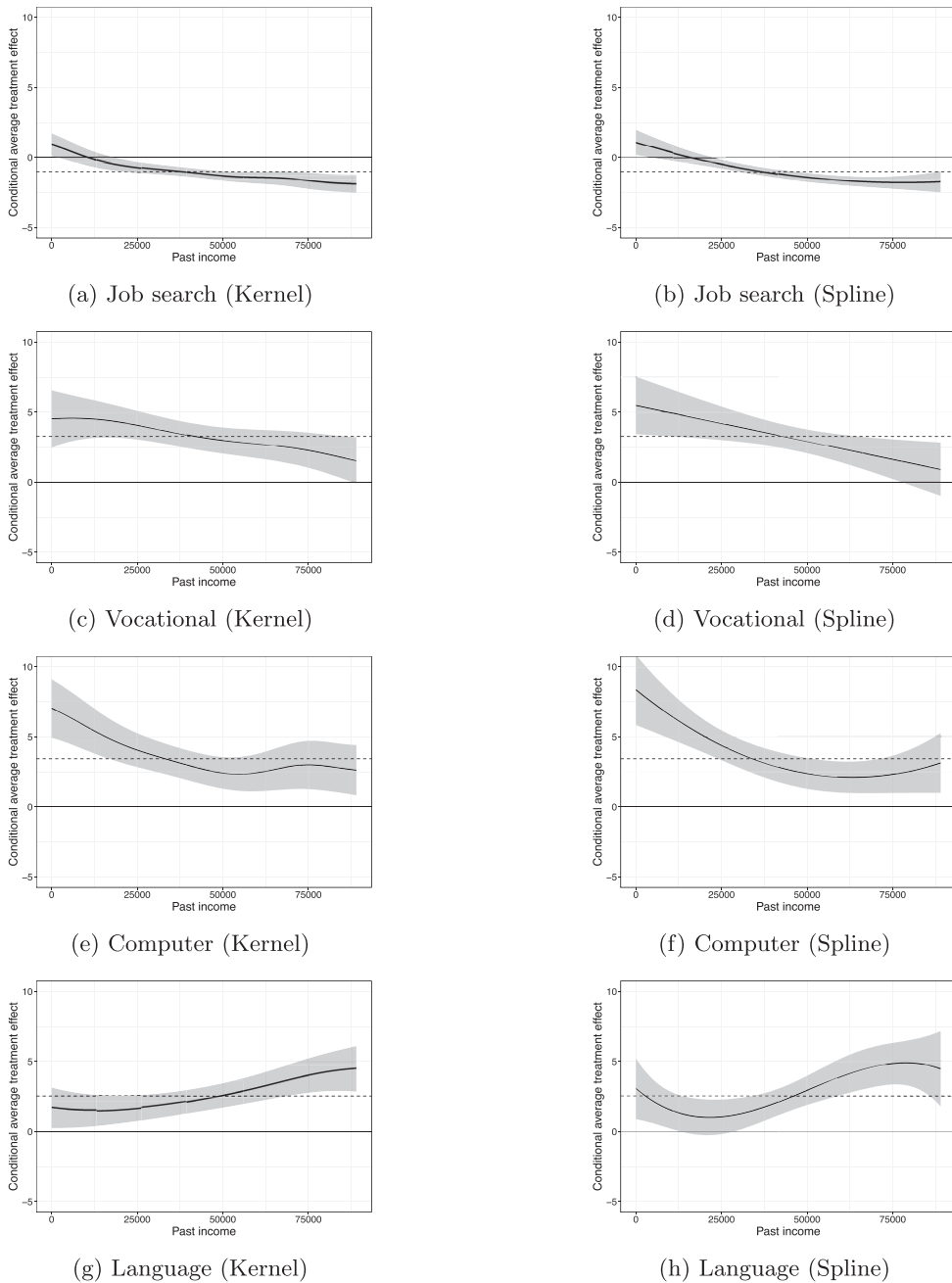


Figure 3. Effect heterogeneity regarding past income. Dotted line indicates point estimate of the respective average treatment effect. Grey area shows 95% confidence interval.

Table 5. Best linear prediction of CATEs.

	Job search (1)	Vocational (2)	Computer (3)	Language (4)
Constant	−0.48 (0.70)	4.46* (2.40)	4.93** (2.37)	6.07*** (2.09)
Female	0.25 (0.27)	−2.12** (0.92)	1.89** (0.90)	−1.61** (0.80)
Age	0.02 (0.01)	0.09* (0.05)	0.05 (0.05)	−0.07 (0.05)
Foreigner	0.50* (0.27)	1.60* (0.90)	−1.20 (0.96)	−2.77*** (0.74)
Medium employability	−0.65* (0.37)	−1.64 (1.18)	−2.36* (1.22)	−0.78 (0.99)
High employability	−1.03** (0.51)	−3.13** (1.55)	−3.15* (1.71)	−0.47 (1.53)
Past income in CHF 10,000	−0.26*** (0.06)	−0.62*** (0.23)	−0.39** (0.19)	0.31* (0.18)
F-statistic	6.95***	4.12***	3.35***	5.22***

Notes: This table shows OLS coefficients and their heteroscedasticity robust standard errors (in parentheses) of regressions run with the pseudo-outcome. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

illustrates that the same strategies that are usually applied to interpret an outcome OLS regression can now be used to interpret the effect OLS regression.

6.2.4. Individualised average treatment effects. We focus on the results based on the out-of-sample variant of the DR- and NDR-learner as the full sample variant leads to severe overfitting with predicted IATEs ranging from −265 to 161 that are up to eight times larger than what is possible given that the outcome is bounded between 0 and 31.²¹ However, Figure 4 shows that the DR-learner produces impossible effect sizes even out-of-sample, which motivates the proposal of the NDR-learner as stabilised variant. Figure 4 provides boxplots of the predicted IATEs and shows several substantial outliers lying below the smallest possible value of −31. However, the descriptive statistics provided in Table S6 and the joint and marginal distributions depicted in Figure S6 of Online Appendix S3.5 document that besides the outliers the distributions are quite similar and correlate with at least 0.90. Not surprisingly, the impact of normalised weights is much larger for the three smaller programmes and nearly negligible for job search programmes. Still, we base the following discussion for all programmes on the more stable results of the NDR-learner.

We conduct a classification analysis as proposed by Chernozhukov et al. (2018) to understand which variables are most predictive of effect sizes. To this end, we split the predicted IATE distributions in quintiles and compare the covariate means of the observations falling into the fifth and first quintile. For comparability, we normalise all covariates to have mean zero and variance one. Table 6 shows the seven variables that have at least one absolute difference between the highest and lowest quintile that is larger than one standard deviation. For example, we observe that the group with the highest effects (the fifth quintile) of a job search programme has a 1.32 standard deviations lower past income compared to the lowest IATE group (the first quintile).

²¹ See Online Appendix S3.5 for results and discussion of the full sample.

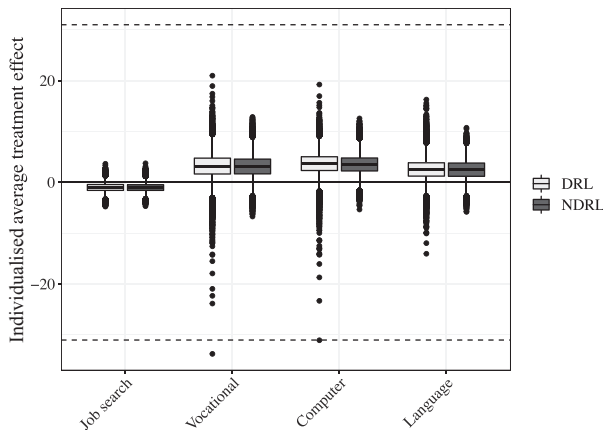


Figure 4. Boxplot of out-of-sample predicted IATEs by DR- and NDR-learner. The figure shows the distribution of IATEs for participating in the programme labelled on the x-axis vs. non-participation estimated by the DR-learner (DRL) and the NDR-learner (NDRL). The dashed line indicates the possible range of the IATE of $[-31, 31]$ to illustrate that several DR-learner estimated IATEs lie outside this bound.

Table 6. Classification analysis of IATEs.

	Job search (1)	Vocational (2)	Computer (3)	Language (4)
Past income	-1.32	-0.84	-1.17	1.01
Previous job: unskilled worker	1.02	0.68	0.34	-1.24
Mother tongue other than German, French, Italian	0.69	0.68	0.00	-1.17
Qualification: some degree	-0.88	-0.65	-0.41	1.15
Swiss citizen	-0.66	-0.60	0.12	1.11
Fraction of months employed last 2 years	-1.06	-0.37	-0.47	0.30
Qualification: unskilled	0.81	0.41	0.32	-1.02

Note: Table shows the differences in means of standardised covariates between the fifth and the first quintile of the respective estimated IATE distribution.

Also the other variables confirm the patterns that we document already in previous subsections. The effects of job search, vocational, and computer training are higher for unskilled workers with lower previous labour market success and foreigners, while the opposite holds for language programmes.²²

6.3. Optimal treatment assignment

The previous section documented substantial heterogeneities in the programme effects. To leverage this heterogeneity for better targeting, we apply the DML-based optimal policy algorithm of Section 3.4. Figure 5(a) shows the simplest decision tree with only one split for the five

²² Table S7 of the Online Appendix shows the classification analysis for all variables.

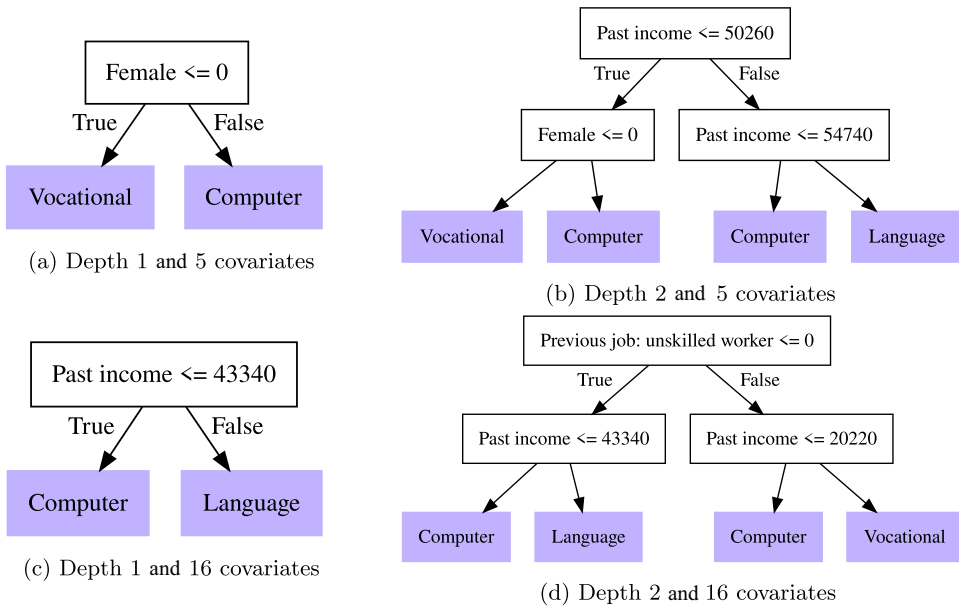


Figure 5. Optimal treatment assignment decision trees of depth two and three. Optimal assignment rules estimated following the procedure defined in Section 3.4.

handpicked covariates. It would allocate men to vocational training and women to computer courses. This split is probably similar to what we would have suggested given the evidence presented in Table 4. For a tree of depth two, such an eyeballing approach has its limits and the algorithmic approach provides a systematic way to arrive at an estimated optimal decision tree. The depth two tree in Figure 5(b) splits first on past income and then recommends to send low earning men to vocational and low earning women to computer training, while high earners (more than CHF55,000) are recommended to participate in language training. In the absence of the possibility to split on gender, the depth one tree in Figure 5(c) splits on past income roughly at the same value where the non-parametric GATEs of computer and language training intersect in Figure 3.²³

Panel A of Table 7 summarises the results of the different trees. It shows the percentage of individuals that are placed in the different programmes. Not surprisingly, all individuals are recommended to be placed into one of the three positively evaluated hard skill enhancing programmes.

One yet unsolved challenge is how to conduct statistical inference about the quality and stability of the decision trees. Athey and Wager (2021) propose a form of cross-validation. To this end, we take the same folds that were used in the cross-fitting procedure to estimate the nuisance parameters. We build the decision tree in four folds and evaluate the value in the left out fold. First, we inspect how often the recommendations based on these trees coincide with the full sample policy rules. Figures S8 and S10 of Online Appendix S3.6 show that the cross-validated trees are not identical to the full sample ones.

²³ Online Appendix S3.6 provides the trees of depth three.

Table 7. Description of estimated optimal policies.

	No programme (1)	Job search (2)	Vocational (3)	Computer (4)	Language (5)
<i>Panel A: Percent allocated to programme</i>					
Depth 1 and 5 variables	0	0	56	44	0
Depth 2 and 5 variables	0	0	32	43	25
Depth 3 and 5 variables	0	0	47	37	17
Depth 1 and 16 variables	0	0	0	54	46
Depth 2 and 16 variables	0	0	23	37	40
Depth 3 and 16 variables	0	0	42	30	27
<i>Panel B: Cross-validated difference to APOs</i>					
Depth 1 and 5 variables	3.75*** (0.40)	4.77*** (0.42)	0.49 (0.44)	0.32 (0.48)	1.21** (0.49)
Depth 2 and 5 variables	3.99*** (0.40)	5.01*** (0.42)	0.73 (0.48)	0.56 (0.46)	1.45*** (0.46)
Depth 3 and 5 variables	3.50*** (0.41)	4.52*** (0.43)	0.24 (0.45)	0.07 (0.49)	0.96** (0.47)
Depth 1 and 16 variables	3.72*** (0.41)	4.73*** (0.42)	0.46 (0.52)	0.29 (0.48)	1.19*** (0.40)
Depth 2 and 16 variables	3.63*** (0.43)	4.65*** (0.44)	0.37 (0.47)	0.20 (0.53)	1.10*** (0.42)
Depth 3 and 16 variables	3.94*** (0.42)	4.96*** (0.43)	0.68 (0.47)	0.51 (0.49)	1.41*** (0.46)

Notes: Panel A shows the percentage of individuals being assigned to a specific programme. Panel B shows a t -test of the difference of the cross-validated policy (standard errors in parentheses) and the APOs of the programmes. * $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

Zhou et al. (2018) propose another validation idea and test whether the optimal policy rules perform significantly better than sending all individuals to the same programme. This is achieved by taking the difference of the APO score of the cross-validated policy rule and the APO score of the programme w : $\hat{\Delta}_{i,w}^{cv}(\pi) = \sum_{t=0}^T \mathbb{1}(\hat{\pi}^{cv}(Z_i) = t) \hat{\Gamma}_{i,t} - \hat{\Gamma}_{i,w}$, where $\hat{\pi}^{cv}(Z_i)$ is the policy rule that is estimated without individual i . A standard t -test on the mean of $\hat{\Delta}_{i,w}^{cv}(\pi)$ tests then whether the cross-validated policy rules are significantly better than sending everybody to the same programme. Note that the cross-validated policy rules do not necessarily coincide with the trees in the full sample and the cross-validation estimates not the value function for that specific tree. This requires to hold out a test set which would be viable for an application with bigger programmes.

The results are provided in Panel B of Table 7. We can interpret the mean of $\hat{\Delta}_{i,w}^{cv}(\pi)$ as average treatment effect comparing a regime under the estimated assignment rule or a regime where everybody is sent to the programme w . This effect is always positive indicating that the estimated rules can leverage the effect heterogeneities to improve the allocation. However, the cross-validated policy rules perform not significantly better than sending just everybody into vocational or computer programmes. This would probably change if we could take costs or capacity constraints into account. However, we do not observe costs in this dataset and the optimal decision tree algorithm is currently not capable of incorporating capacity constraints in a

systematic way. We leave both extensions for future research using a more detailed database on both costs and capacity constraints.

7. DISCUSSION AND CONCLUSIONS

This paper considers recent methodological developments based on double machine learning (DML) through the lens of a standard programme evaluation under unconfoundedness. DML-based methods provide a convenient toolbox for a comprehensive programme evaluation as different parameters of interest can be estimated using the same framework and a combination of standard statistical software. The application to an Active Labour Market Policy evaluation shows that the methods also produce plausible results in practice. The only exception is the DR-learner that required a modification, the newly introduced NDR-learner, before producing stable results for all individualised treatment effects. However, several conceptual and implementation issues remain open for investigation and refinement.

In general, we know little about how to choose the estimator for the nuisance parameters. The pool of potential machine learning algorithms and their combinations is large and little is known, e.g., about the trade-off between high prediction performance and computation time in the causal setting. Also clear recommendations for the implementation of cross-fitting are missing. Another open question is how to deal with common support in general and for each estimand specifically. The literature on trimming rules is well developed for propensity score-based methods estimating average effects. However, we are not only interested in average effects and the propensity score is not the only nuisance parameter of DML. It remains an open question whether the established trimming methods are also sensible for DML when common support becomes an issue.

The estimators for flexible heterogeneous treatment effects provide interesting new tools. However, it is currently not clear to what extent we can actually explore heterogeneity or to what extent we need to pre-define the heterogeneity of interest. The possibility to summarise pre-defined heterogeneity of interest using OLS, kernel or series regressions provide clearly valuable and easy to use options in applications. The instability of methods that aim for individualised heterogeneous effects shows that they should be used with caution and more research is required to investigate whether adjustments like the proposed NDR-learner are useful beyond the application of this paper.

The estimation of optimal treatment assignment rules is mostly unexplored in practice and many interesting issues in applications regarding inference, the implementation of different constraints, more flexible rules than decision trees, or the choice of variables that could or should enter the set of policy variables, which could be explored in future research.

The investigation of these DML specific questions, but also the comparison with other more specialised causal machine learning methods for each estimand, provides another interesting direction of future research. Such evidence would help to understand and guide which choices are critical in applications similar to the one in this paper.

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